Predicting 30-Day Hospital Readmission in Diabetic Patients: A Comparative Machine Learning Study

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Abstract (This is the Abstract Head style)

Hospital readmission within 30 days is an important healthcare quality indicator, especially for patients with diabetes, who often present with complex comorbidities, frequent healthcare utilization, and elevated risk of preventable return visits. In this project, I addressed the problem of predicting 30-day hospital readmission for diabetic patients using supervised machine learning on the UCI Diabetes 130-US Hospitals dataset. I formulated the task as a binary classification problem, where the positive class corresponds to readmission within 30 days and the negative class corresponds to all other encounters.

Our workflow included data cleaning, target construction, feature engineering, train/test splitting, preprocessing for mixed numerical and categorical inputs, and imbalance handling with oversampling inside a pipeline. I compared six baseline models: Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, Gradient Boosting, and a Multi-Layer Perceptron. I evaluated models using stratified 5-fold cross-validation and a holdout test set, with particular emphasis on AUC, precision, recall, and F1-score due to strong class imbalance.

Among the baseline models, Gradient Boosting performed best overall on the holdout set, achieving the highest test AUC (0.678) and the highest recall (0.630), although with low precision (0.178). Random Forest achieved high accuracy (0.889) but extremely low recall (0.027), demonstrating that accuracy alone is misleading for this imbalanced task. Hyperparameter tuning improved Random Forest modestly, increasing test AUC from 0.665 to 0.672, but it still did not surpass Gradient Boosting. Feature importance analysis showed that prior inpatient utilization, discharge disposition, number of laboratory procedures, number of medications, and time in hospital were among the most influential predictors. Overall, the project demonstrates that clinically plausible readmission risk signals can be extracted from structured EHR-style data, while also highlighting the limitations imposed by imbalance, noisy tabular features, and moderate predictive separability.

Code — https://github.com/JacquiLJQ/eece5644-final-project

Introduction

Hospital readmissions within 30 days are widely used as a marker of care quality, continuity, and discharge effectiveness. For patients with diabetes, this problem is particularly important because diabetes is associated with frequent hospital use, multiple complications, and substantial healthcare costs. Predicting which diabetic patients are most likely to be readmitted soon after discharge could support earlier intervention, better discharge planning, and more targeted transitional care.

This task is interesting from both a machine learning and healthcare perspective. From a healthcare standpoint, a useful prediction model could help hospitals prioritize scarce follow-up resources toward patients at highest risk. From a modeling standpoint, the task is challenging because hospital readmission is influenced by many interacting factors, including disease burden, prior utilization, medication complexity, discharge context, and broader social determinants that may not be fully captured in the available data.

Three challenges motivated this project. First, the outcome is strongly imbalanced: relatively few patients are readmitted within 30 days, so a naïve model can achieve deceptively high accuracy by predicting the majority class. Second, the dataset is high-dimensional and heterogeneous, containing a mix of numerical variables, many categorical variables, and several columns with large numbers of levels. Third, the data contain quality issues such as placeholder-coded missingness and sparse categories, which complicate preprocessing and model interpretation. Because of these challenges, this project emphasizes not just prediction, but also careful preprocessing, appropriate evaluation, and clinically meaningful interpretation.

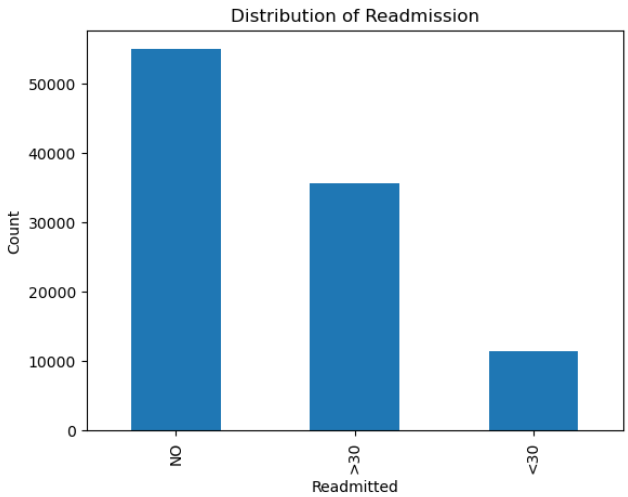

Figure 1: Distribution of Readmission

Background

This project uses supervised machine learning for binary classification. The goal is to estimate, for each hospital encounter, the probability that the patient will be readmitted within 30 days. Because the target is imbalanced, the most important evaluation metrics are not accuracy alone, but AUC, precision, recall, and F1-score.AUC (Area Under the ROC Curve) measures how well a model ranks positive examples above negative ones across thresholds. It is especially useful when comparing models independently of a single classification cutoff. Precision measures the fraction of predicted positives that are truly positive. Recall measures the fraction of actual positives that are successfully identified. F1-score is the harmonic mean of precision and recall and is useful when the positive class is relatively rare.

We compared six commonly used algorithms. Logistic Regression provides a simple linear baseline with probabilistic outputs. KNN offers a nonparametric comparison point but can struggle in high-dimensional mixed-feature spaces. Decision Tree provides an interpretable nonlinear baseline but is prone to overfitting. Random Forest reduces tree variance by averaging across many trees and often performs well on tabular data. Gradient Boosting builds trees sequentially to correct previous errors and often yields strong discrimination on structured datasets. MLP (Neural Net) provides a simple deep learning baseline, although dense neural networks are not always optimal for modest-size mixed tabular datasets without extensive tuning.

Because the data contains both numerical and categorical variables, preprocessing is critical. Numerical features were imputed and scaled, while categorical variables were imputed and one-hot encoded. In addition, the class imbalance required explicit handling within the modeling pipeline to avoid biased training behavior.

Related Work

Hospital readmission prediction, especially for patients with diabetes, has been widely studied in the machine learning and healthcare literature. Because readmission is both clinically important and operationally costly, researchers have explored a broad range of approaches, including classical statistical models, standard machine learning methods, ensemble tree models, explainable AI frameworks, and more recently, deep learning architectures tailored to heterogeneous clinical data. This broad literature shows that the task is well established, but also challenging, since performance depends heavily on preprocessing choices, validation design, class imbalance handling, and the ability to capture complex interactions in structured health records.

In this report, I focus on two recent papers that provide useful comparison points for our project. The first, by Emi-Johnson and Nkrumah, represents a strong modern baseline using standard machine learning methods on the same UCI diabetes readmission dataset. They compared logistic regression, random forest, XGBoost, and deep neural networks, and reported that XGBoost achieved the best AUC-ROC of 0.667. Their interpretation results further showed that prior admissions, number of medications, diabetes medication usage, and admission-related variables were among the most important predictors.

The second paper, by García-Mosquera et al., represents a more innovative modeling direction. They proposed a Transformer-based architecture for diabetes readmission prediction and argued that this approach can integrate heterogeneous feature types and model feature interactions effectively. Their study used patient-wise cross-validation and positioned the Transformer as a more advanced alternative to conventional tabular machine learning models.

These two studies are useful because they bracket the space in which our project sits. The Emi-Johnson and Nkrumah paper reflects a strong benchmark-oriented machine learning approach, while the Transformer paper reflects a higher-complexity neural architecture designed to push performance further. By contrast, our project is intentionally more conservative and classroom oriented. Rather than optimizing one highly specialized model, I built a transparent, reproducible pipeline that compares several standard algorithms under the same preprocessing and validation framework, incorporates oversampling to address imbalance, and uses feature importance plus descriptive plots for interpretation.

This design choice also explains why I did not use either XGBoost or a Transformer in the present project. XGBoost is widely recognized as a strong method for tabular data, but it often benefits from more extensive hyperparameter tuning and model-specific optimization than was practical within the scope of this report. In addition, our experiments already

included Gradient Boosting and Random Forest, which allowed us to study the broader behavior of tree-based ensemble methods without adding another heavily tuned boosting framework. Likewise, a Transformer-based approach would require greater modeling complexity, more computational effort, and a more specialized design process, which would have shifted the project away from its main goal: establishing a clear and reproducible benchmark pipeline for this dataset.

Our findings are broadly consistent with the literature. Ensemble tree-based methods were among the strongest performers in our experiments, which aligns with prior work showing that boosting and related ensemble approaches are effective for this problem. In addition, clinically intuitive utilization-related variables emerged as important predictors in both our notebook and the published studies. At the same time, our best AUC remained moderate rather than state-of-the-art. This is not surprising, since I did not use XGBoost, Transformer architectures, patient-wise validation, or extensive model-specific optimization. Instead, our contribution is a transparent end-to-end comparison showing what can be achieved with a carefully constructed but methodologically simpler pipeline.

Problem Formulation and Dataset

We formulate the task as a binary classification problem. The original readmitted variable in the dataset has three categories: NO, >30, and <30. In this project,I define the positive class as readmission within 30 days (<30) and the negative class as all other outcomes (NO and >30). This definition focuses the model on early readmission, which is the most clinically actionable outcome.

The dataset contains 101,766 encounters and originally 50 columns. Initial inspection showed a mixture of demographic, admission, discharge, utilization, diagnosis, and medication-related variables. The raw target distribution was highly imbalanced: 54,864 encounters were labeled NO, 35,545 were labeled >30, and only 11,357 were labeled <30. After binary target construction, the positive class accounted for approximately 11.16% of the data, confirming a strong imbalance problem.

Exploratory analysis also highlighted the high-dimensional and mixed-type structure of the data. The dataset contained 49 input features prior to target construction, including a large number of categorical variables. Some categorical columns also exhibited high cardinality, such as diagnosis-related fields. In addition, the raw data used placeholder-style missing coding such as ?, which required cleaning before modeling.

During preprocessing, columns with more than 80% missingness were dropped. In practice, this led to the removal of weight, max_glu_serum, and A1Cresult. The cleaned dataset had 45 columns, after which feature engineering produced a final modeling table with 50 predictors. Feature engineering added several derived variables, including counts of diabetes medications used, counts of diabetes medications changed, total service utilization, and an indicator for prior inpatient use. After cleaning and feature construction, the data were split into training and test sets, preserving the outcome proportion in each split. The final split yielded 81,412 training rows and 20,354 test rows, with nearly identical positive rates in train and test.

Methodology

Our methodology follows a standardized supervised learning pipeline for structured tabular healthcare data. The overall workflow includes preprocessing, class imbalance handling, model fitting, cross-validation, holdout evaluation, and limited hyperparameter tuning for selected models.

Preprocessing pipeline

Because the dataset contains both numeric and categorical variables, I use a column-wise preprocessing strategy. Numeric variables are imputed using the median and then standardized. Categorical variables are imputed using the most frequent category and one-hot encoded. This preprocessing is implemented through a ColumnTransformer, allowing the two variable types to be handled consistently within the same modeling framework.

To address class imbalance, each model is wrapped in an ImbPipeline from the imblearn library. Each pipeline contains three sequential stages:

- Preprocessor: transforms raw numeric and categorical features
- Sampler: applies RandomOverSampler to the training data
- Model: fits the classifier

This design is important because it ensures that oversampling occurs inside the training workflow rather than before data splitting or outside cross-validation. As a result, oversampling is applied only to the training portion of each fold, which reduces the risk of information leakage and produces a fairer estimate of model performance.

Model Compared

We compare six classification models representing a range of inductive biases and complexity levels:

- Logistic Regression
- K-Nearest Neighbors
- Decision Tree
- Random Forest
- Gradient Boosting
- Neural Network (MLP)

These models were selected to provide a broad benchmark across linear, instance-based, tree-based, ensemble, and

neural-network methods. Logistic Regression serves as an interpretable baseline. KNN offers a local distance-based approach, though it may be challenged by high-dimensional sparse features after one-hot encoding. Decision Trees capture nonlinear decision rules but can overfit. Random Forest and Gradient Boosting provide more powerful ensemble-based alternatives for tabular data. MLP serves as a neural baseline for comparison.

In the baseline experiments, the model settings were chosen as follows: Logistic Regression used max_iter=2000; KNN used n_neighbors=25; Random Forest used 300 trees; Gradient Boosting and Decision Tree used standard scikit-learn settings with a fixed random seed; and the MLP used two hidden layers with sizes (64, 32) and ReLU activation.

Validation strategy

We evaluate models using 5-fold stratified cross-validation on the training set. Stratification preserves the minority-class proportion in each fold and is therefore appropriate for this imbalanced classification problem. For each fold, I compute:

- ROC-AUC
- Accuracy
- Precision
- Recall
- F1-score

Among these, ROC-AUC is treated as the primary model-selection metric because it measures ranking performance across thresholds and is less sensitive to a single fixed classification cutoff. Precision, recall, and F1-score are also reported because they provide additional insight into minority-class detection performance.

After cross-validation, each model is fit on the full training set and evaluated on a separate holdout test set. This gives an out-of-sample comparison of final model performance and allows us to identify the strongest baseline model under the same preprocessing and resampling framework.

Hyperparameter tuning

After the baseline comparison, I performed additional tuning for Random Forest using a randomized search proceedure. The tuned Random Forest was then compared with the untuned version on the holdout set. This analysis allows us to assess whether modest hyperparameter optimization can materially improve performance beyond the baseline configuration.

Model interpretation

To improve interpretability, I also examined feature importance from the tuned Random Forest model. Because the preprocessing pipeline includes one-hot encoding, feature names were extracted from the transformed design matrix before importance values were computed.

Empirical Results and Evaluation

The cross-validation results showed a clear pattern. Gradient Boosting achieved the highest mean cross-validated AUC (0.669) and also the highest recall (0.601) among the baseline models. Random Forest had a slightly lower mean AUC (0.651) but extremely high mean accuracy (0.888) and very low recall (0.018), indicating that it was dominated by majority-class predictions. Logistic Regression performed moderately with a mean AUC of 0.628, while KNN, MLP, and Decision Tree lagged behind.

model	cv AUC	cv accuracy	cv precision	cv recall	cv f1
Gradient Boosting	0.669	0.638	0.175	0.601	0.271
Random Forest	0.651	0.888	0.501	0.018	0.034
Logistic Regression	0.628	0.639	0.161	0.532	0.248
KNN	0.583	0.578	0.138	0.533	0.220
MLP	0.569	0.805	0.156	0.169	0.162
Decision Tree	0.542	0.793	0.169	0.219	0.191

Chart 1: Cross-Validation Result of All Models

model	test AUC	test accuracy	test precision	test recall	test f1
Gradient Boosting	0.678	0.635	0.178	0.630	0.278
Random Forest	0.665	0.889	0.592	0.027	0.051
Logistic Regression	0.643	0.637	0.165	0.556	0.255

model	test AUC	test accuracy	test precision	test recall	test f1
KNN	0.604	0.586	0.144	0.547	0.228
MLP	0.581	0.818	0.164	0.155	0.160
Decision Tree	0.548	0.795	0.177	0.229	0.200

Chart 2: Test Result of All Models

This pattern persisted on the holdout test set. Gradient Boosting again performed best overall, with test AUC = 0.678, accuracy = 0.635, precision = 0.178, recall = 0.630, and F1 = 0.278. Logistic Regression ranked next in AUC at 0.643, followed by Random Forest at 0.665 if ordered by AUC it is actually second, but its recall remained extremely low (0.027), limiting its usefulness as a screening model. KNN and MLP both produced weaker discrimination, and the single Decision Tree performed worst overall.

The comparison between Gradient Boosting and Random Forest illustrates why accuracy alone is not appropriate for this task. Random Forest's holdout accuracy of 0.889 looks superficially excellent, but its recall of 0.027 means it missed almost all true 30-day readmissions. Gradient Boosting, by contrast, accepted many more false positives in exchange for substantially higher sensitivity to the minority class. In a healthcare screening context, this tradeoff is often more reasonable because failing to identify high-risk patients may be more costly than flagging some additional low-risk patients.

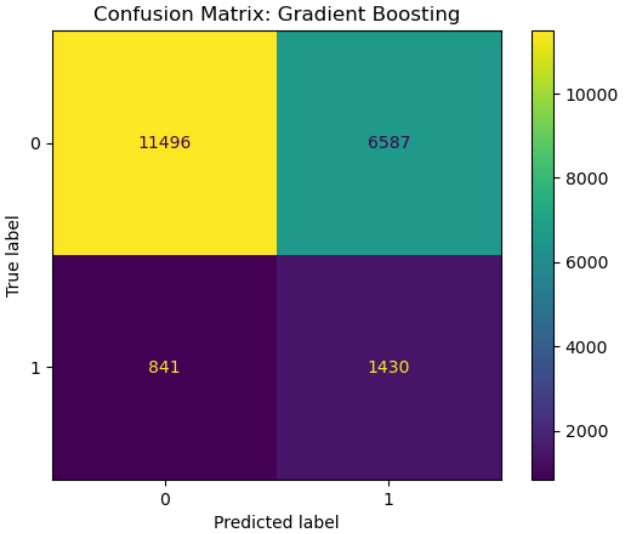

Figure 2: Confusion Matrix for Gradient Boosting

The confusion matrix for Gradient Boosting makes this tradeoff concrete. The model correctly identified 1,430 true positives and 11,496 true negatives, but it also produced 6,587 false positives and 841 false negatives. This means the model captured about 63% of true readmissions while generating many false alarms, which explains its relatively high recall but low precision. In practical terms, the model behaves like a broad screening tool: it is willing to over-flag patients to avoid missing too many high-risk cases.

Hyperparameter tuning modestly improved Random Forest. The untuned model achieved test AUC = 0.665, while the tuned version reached 0.672. This confirms that tuning helped, but only slightly. More importantly, even after tuning, Random Forest still did not surpass Gradient Boosting's holdout AUC. This suggests that the main limitations are not just hyperparameter choices, but also the intrinsic difficulty of the task, the class imbalance, and the restricted predictive signal available in the structured features.

	version	test_auc
0	Untuned Random Forest	0.664649
1	Tuned Random Forest	0.672191

Chart 3: Test AUC of Random Forest Before and After Tunning

The interpretation results were clinically plausible. The most important variables in the tuned Random Forest included number of inpatient visits, discharge disposition ID, number of lab procedures, number of medications, time in hospital, and number of diagnoses. These variables all reflect healthcare utilization, disease burden, and care complexity. The post-hoc plots reinforced this interpretation: patients with more prior inpatient visits tended to have higher observed readmission rates, and readmission rates differed substantially across discharge disposition categories. Together, these findings suggest that previous utilization and discharge context are among the strongest signals available in this dataset.

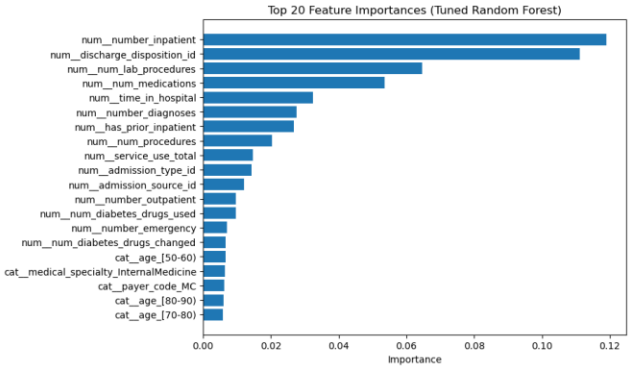

Figure 3: Top-20 Feature Importance Values

What worked well in this project was the pipeline-based design: preprocessing, resampling, and modeling were applied consistently, and the workflow made it easy to compare multiple algorithms fairly. The interpretation results were also coherent and aligned with clinical intuition. What did not work as well was the ability to achieve both high recall and high precision simultaneously. The best model still produced many false positives, and the overall AUC remained moderate. With more time, improvements would likely come from threshold tuning, class-weighted learning, more advanced gradient-boosted tree models such as XGBoost or LightGBM, probability calibration, and fairness-aware subgroup evaluation.

Compared with the Emi-Johnson and Nkrumah study, our best holdout AUC of 0.678 is very close to the 0.667 AUC-ROC they reported for their strongest model, XGBoost. Compared with the Transformer paper, our results are likely weaker, since García-Mosquera et al. describe their model as achieving slightly better performance than previous approaches on the same dataset under patient-wise cross-validation. This difference is reasonable because our project prioritized reproducibility and breadth of baseline comparison over specialized model design.

Real-World Impact

A model of this kind could be useful as a decision-support tool for hospitals, discharge planning teams, and care coordinators. By identifying patients at elevated risk of 30-day readmission, hospitals could prioritize follow-up calls, medication review, diabetes education, or enhanced transitional care for patients who may benefit the most. Prior work also emphasizes that reducing diabetes-related readmissions has major cost implications and could support more efficient use of hospital resources.

At the same time, such a model should not be treated as a stand-alone decision-maker. Readmission is influenced not only by clinical characteristics, but also by access to follow-up care, insurance coverage, transportation, housing stability, and other social factors that are only partially represented in structured encounter-level data. A model trained on historical records may therefore reflect structural inequities present in the healthcare system rather than purely clinical need.

There are also fairness concerns. Variables such as payer code, discharge-related variables, and utilization history may encode social disadvantage or prior access differences. If predictions are used uncritically, the model could amplify inequitable allocation of resources or misclassify some demographic groups more often than others. These risks are especially relevant in healthcare, where false negatives may deprive vulnerable patients of helpful support, while false positives may divert limited resources away from others.

For these reasons, any real-world deployment would require more than good aggregate performance. It would require subgroup-level evaluation, calibration checks, threshold selection aligned with operational goals, and human oversight. In that sense, the present project is best viewed as an exploratory but practically relevant demonstration of how machine learning can support readmission risk stratification rather than a deployment-ready clinical product.

Conclusion and Future Work

In this project,I built a complete machine learning pipeline to predict 30-day hospital readmission among diabetic patients using the UCI Diabetes 130-US hospitals dataset. The task was formulated as a binary classification problem, with readmission within 30 days treated as the positive class. Because the dataset contains mixed data types and substantial class imbalance, I used a preprocessing-and-resampling workflow implemented through an ImbPipeline, combining column-wise preprocessing, RandomOverSampler, and model fitting in a single framework.

Our experiments compared Logistic Regression, KNN, Decision Tree, Random Forest, Gradient Boosting, and MLP. Across both cross-validation and holdout evaluation, Gradient Boosting emerged as the strongest baseline model, achieving the best holdout AUC (0.678). Hyperparameter tuning improved Random Forest modestly, from 0.665 to 0.672 AUC, but it still did not outperform Gradient Boosting. Feature-importance analysis further suggested that prior inpatient utilization, discharge disposition, treatment intensity, and hospital stay characteristics were among the most influential predictors.

Several lessons emerged from this project. First, class imbalance must be handled carefully, and model comparison should rely on AUC, precision, recall, and F1-score rather than accuracy alone. Second, tree-based ensemble models appear especially effective for structured hospital tabular data, outperforming both simpler linear baselines and the neural network baseline tested here. Third, even with a careful pipeline, the overall discrimination remains moderate, which indicates that 30-day readmission is a difficult outcome to predict from structured EHR features alone.

There are several directions for future work. One natural extension would be to compare RandomOverSampler with other imbalance-handling methods such as SMOTE or cost-sensitive learning. Another would be to perform broader hyperparameter tuning across more models, especially Gradient Boosting or more modern boosting methods such as XGBoost. Additional improvements could come from richer feature engineering, including diagnosis aggregation, medication-category summaries, or temporal utilization features. It would also be valuable to evaluate calibration and sub-

group performance across age, sex, race, or insurance-related variables to better understand fairness and practical deployment concerns. More generally, future versions of this project could benefit from incorporating social determinants of health or longitudinal information, since readmission risk is influenced by many factors not fully represented in a single structured encounter-level record.

If I were to continue this project with more time, I would focus on three improvements: stronger boosting models, more clinically meaningful feature engineering, and fairness-oriented evaluation. I would also compare threshold-tuning strategies, since a hospital may prefer higher recall if the goal is to identify as many at-risk patients as possible. For future students working on a similar project, my main advice would be to build a clean and leakage-free pipeline first, choose evaluation metrics that match the clinical objective, and interpret results carefully rather than relying only on the single highest accuracy number.

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